

POWER + MOTION

ZRCore

Power, motion, and expansion core for robotics

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ZRCore

Board

Motor, power, and expansion core for ZROne

IP5306 Power Management / Charging

USB charging, power-path management, and protection for mobile applications.

Interconnect / Expansion

Board-to-board headers for stacking with ZROne or custom expansion boards. Daisy chain boards or add your own I2C sensors or displays on one of the multiple I2C connections.

Headers for Servo Motors

12 connections (J1-J6) for servo motors, PWM output to electronic speed controllers or dimmable LEDs.

Dual 18650 Battery Bay

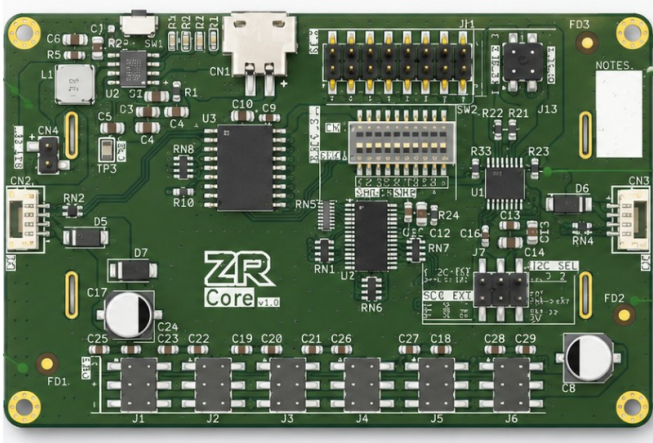
Portable power on the opposite side with capacity for two 18650 cells. Provides over 5 hours of run time. Build your own docking station and have a always on platform.

8 GPIO MCP23008

8 input/output lines using the robust and simple MCP23008 chip

Dual H-Bridge Motor Driver

Drive two DC motors (like the N20) or a single stepper motor from the DRV8833 dual H-bridge driver, output on J13



CONTROL

PCA9685 PWM

16 channels of 12-bit PWM for servos, LEDs, and actuators.

EXPANSION

MCP23008 GPIO

8-bit I/O expander over I2C for sensors and peripherals.

MOTION

DRV8833 Driver

Dual H-bridge driver for two DC motors with current control and protection.

PWM Control

16-channel, 12-bit

GPIO

8x over I2C

Power

IP5306 path

Motor Drive

Dual H-bridge

Expansion

Stack headers

Build the robot, not the electronics. ZRCore brings battery management, motor control, high-channel PWM, GPIO expansion, and I²C interconnect into one modular robotics board.

POWER

Dual 18650 battery bay with IP5306 USB charging, power-path management, and battery protection.

MOTION

DRV8833 dual H-bridge for two DC motors or one stepper motor. External motor supply supported up to **10.8V maximum**.

CONTROL

PCA9685 provides 16 channels of 12-bit PWM for servos, ESCs, LEDs, and other actuators.

EXPANSION

MCP23008 adds 8 GPIO over I²C, while edge connectors support board-to-board I²C and power pass-through.

Designed for the ZoneRobotics ZROne platform, but useful as a standalone foundation for custom robots, rovers, hexapods, robotic arms, and other actuator-heavy projects.

ZRCore v1.1 — Technical Overview

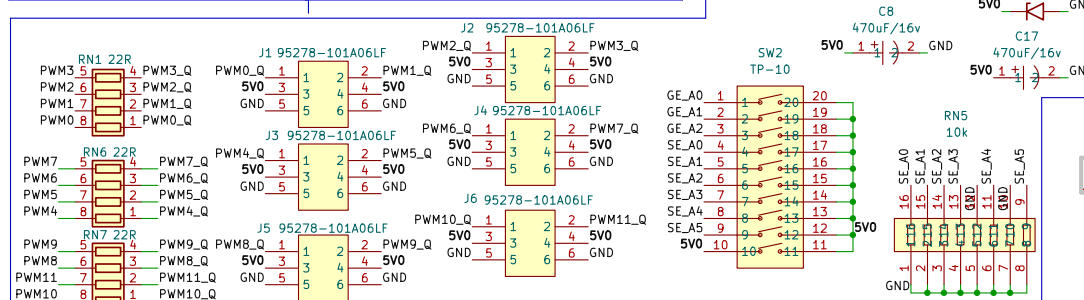
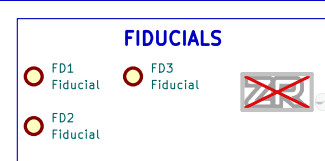
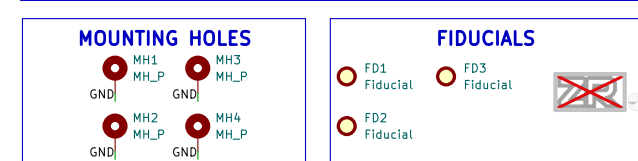
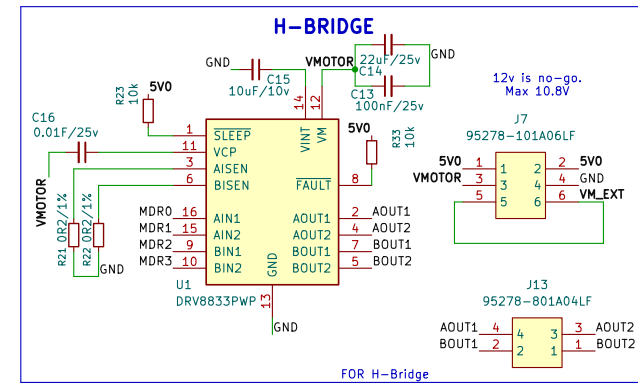
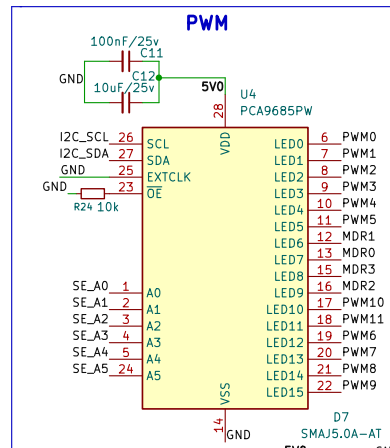
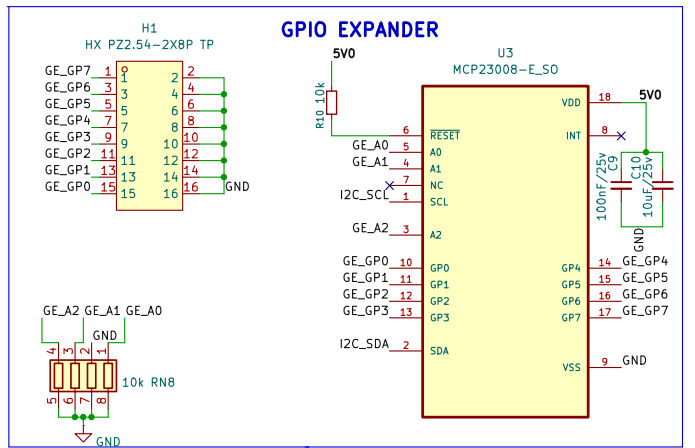
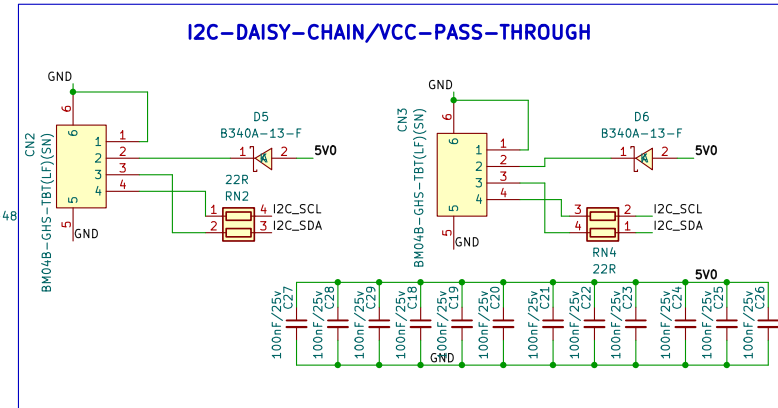
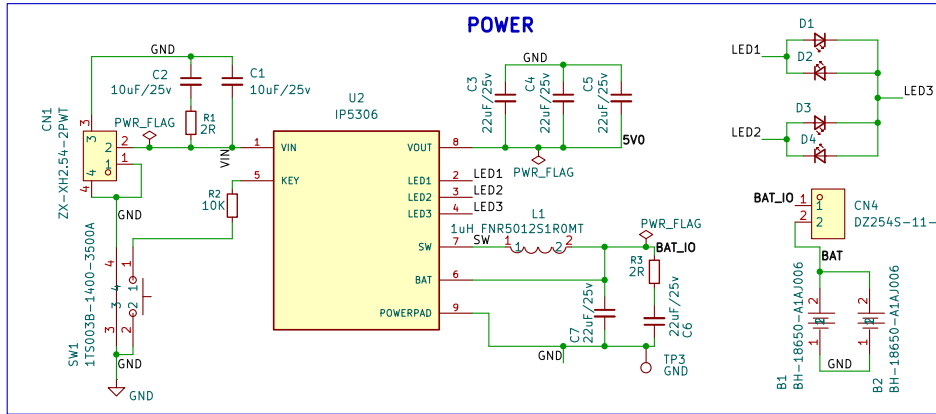
ZRCore is designed as the power, motion, and expansion layer of a modular robotics system. The main controller communicates with the onboard peripheral ICs over I²C, while dedicated connectors keep motors, servos, power, and expansion wiring organized.

FUNCTION	DETAIL
Power management	IP5306 • USB charging • power-path management • battery protection
Battery	2 × 18650 lithium-ion cells in the onboard battery bay
PWM controller	PCA9685 • 16 channels • 12-bit PWM
PWM connections	J1–J6 • 12 dedicated 3-wire outputs for servos, ESCs, or dimmable LEDs
Motor driver	DRV8833 dual H-bridge • 2 DC motors or 1 stepper motor
Motor power	External motor supply • 10.8V maximum
GPIO expansion	MCP23008 • 8 GPIO over I ² C
I ² C expansion	Edge connectors for board-to-board I ² C, sensors, displays, and expansion boards
Configuration	SW2 DIP switch for I ² C address / GPIO expansion configuration

IMPORTANT POWER LIMIT

10.8V maximum motor voltage. Do not exceed 10.8V on the external motor-power input. Follow the documentation and schematic when connecting batteries, motors, and external power.

REFERENCE SCHEMATIC



ZRCore v1.1 — Connector & I/O Reference

Use this table as a quick wiring reference. Voltage ranges below describe the electrical domains shown in the supplied schematic. Where the schematic does not define a connector current rating or absolute voltage limit, that is explicitly noted rather than inferred.

CONNECTOR	TYPE	SIGNALS / PINS	NOMINAL / LIMIT		PURPOSE
CN1	Power input	VIN, GND	5V nominal		Feeds the IP5306 power-management/charging stage.
CN4	External battery	BAT, GND	Battery rail		External battery connection; the onboard two 18650 cells connect to the same BAT/GND rail.
CN2	I ² C / pass-through	I2C_SCL, I2C_SDA, 5V, GND	0–5V logic / 5V rail		Edge connector for I ² C daisy-chaining and power pass-through.
CN3	I ² C / pass-through	I2C_SCL, I2C_SDA, 5V, GND	0–5V logic / 5V rail		Second edge connector for continuing the I ² C bus and 5V/GND.
J1	PWM outputs	PWM0_Q, PWM1_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J2	PWM outputs	PWM2_Q, PWM3_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J3	PWM outputs	PWM4_Q, PWM5_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J4	PWM outputs	PWM6_Q, PWM7_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J5	PWM outputs	PWM8_Q, PWM9_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J6	PWM outputs	PWM10_Q, PWM11_Q, 5V, GND	PWM: 0–5V logic; 5V supply		Two PWM channels for servo/ESC/LED connections.
J7	Motor/power interface	VMOTOR, 5V, GND, VM_EXT	VMOTOR: up to 10.8V max; 5V rail		Provides motor supply/interface connections for the DRV8833 stage and external motor power.
J13	Motor outputs	AOUT1, AOUT2, BOUT1, BOUT2	Motor output follows VMOTOR; max motor supply 10.8V		Four H-bridge outputs for two DC motors or one stepper motor.
H1	GPIO expansion	GE_GP0–GE_GP7, GND	0–5V logic domain		Eight GPIO lines from the MCP23008 I/O expander.
SW2	Configuration	GE_A0–GE_A2, SE_A0–SE_A5, 5V/GND	5V logic domain		DIP-switch configuration for GPIO-expander and PWM/I ² C address selection.

Connector-current note: The supplied schematic identifies the signal and power rails but does not specify connector-specific current ratings. Do not treat the absence of a stated limit as permission to exceed the ratings of the underlying ICs, connectors, wiring, batteries, or attached loads. Refer to the component datasheets and test conditions for final electrical limits.